

$$a) \int \left[\left(x - \frac{1}{x} \right) \left(x - \frac{1}{x} \right) \right] dx$$

$$\int \left(x^2 - 1 - 1 + \frac{1}{x^2} \right) dx$$

$$\int x^2 dx - \int 2 dx + \int x^{-2} dx$$

$$\frac{x^3}{3} - 2x - \frac{1}{x} + C$$

$$c) \int 3x^2 - 4x dx$$

$$3 \int x^2 - 4 \int x dx$$

$$x^3 - 2x^2 + C$$

$$b) \int \left(3\sqrt{x} + \frac{1}{\sqrt{x}} \right) dx$$

$$3 \int x^{\frac{1}{2}} dx + \int x^{-\frac{1}{2}} dx$$

$$2x^{\frac{3}{2}} + \frac{x^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$2x^{3/2} + 2x^{1/2} + C$$

$$d) \int 6x^2 + 8x - 15x - 20 dx$$

$$6 \int x^2 dx - 7 \int x dx - \int 20 dx$$

$$2x^3 - 7 \frac{x^2}{2} - 20x + C$$

$$e) \int \left[\frac{4x-6}{\sqrt[3]{x}} \right] dx$$

$$4 \int \frac{x}{\sqrt[3]{x}} dx - \int \frac{6}{\sqrt[3]{x}} dx$$

$$4 \int x^{\frac{2}{3}} dx - 6 \int x^{-\frac{1}{3}} dx$$

$$4 \frac{x^{\frac{5}{3}}}{\frac{5}{3}} - 6 \frac{x^{\frac{2}{3}}}{\frac{2}{3}} + C$$

$$12 \frac{x^{\frac{5}{3}}}{5} - 9 x^{\frac{2}{3}} + C$$

$$f) \int \frac{3x}{\sqrt[3]{x^2+4}}$$

$$u = x^2 + 4 \quad \frac{du}{dx} = 2x \quad \frac{du}{2} = x dx$$

$$\frac{3}{2} \int \frac{1}{\sqrt[3]{u}} du$$

$$\frac{3}{2} \int u^{-\frac{1}{3}} du$$

$$\frac{9}{4} u^{\frac{2}{3}} + C$$

$$\frac{9}{4} (x^2+4)^{\frac{2}{3}} + C$$

$$g) \int \frac{(\sqrt{x}+4)^2}{x^3} dx$$

$$\int \frac{x+8\sqrt{x}+16}{x^3} dx$$

$$\int \frac{x}{x^3} dx + 8 \int \frac{\sqrt{x}}{x^3} dx + 16 \int \frac{1}{x^3} dx$$

$$\int x^{-2} dx + 8 \int x^{-\frac{5}{2}} dx + 16 \int x^{-3} dx$$

$$-x^{-1} - 16 \frac{x^{-\frac{3}{2}}}{\frac{3}{2}} - 8x^{-2} + C$$

$$h) \int \left(1 + \frac{1}{x}\right)^{-3} \cdot \frac{1}{x^2} dx$$

$$u = 1 + \frac{1}{x} \quad \frac{du}{dx} = -\frac{1}{x^2} \quad -du = \frac{1}{x^2} dx$$

$$-\int u^{-3} du$$

$$-\frac{u^{-2}}{-2} + C$$

$$+ \frac{\left(1 + \frac{1}{x}\right)^{-2}}{2} + C$$

$$i) 8 \int \cos\left(\frac{x}{2}\right) dx$$

$$u = \frac{x}{2} \quad \frac{du}{dx} = \frac{1}{2} \quad 2du = dx$$

$$8 \cdot 2 \int \cos(u) du$$

$$16 \cdot \sin\left(\frac{x}{2}\right) + C$$

$$J) \int (\cos(x) - \sin(x))(\cos(x) - \sin(x)) dx$$

$$\int 1 - \sin(2x) dx$$

$$\int 1 dx - \int \sin(2x) dx$$

$$u = 2x \quad \frac{du}{dx} = 2 \quad \frac{du}{2} = dx$$

$$x - \frac{1}{2} \int \sin(u) du$$

$$x + \frac{\cos(2x)}{2} + C$$

$$k) \int x^2 [\sin(4x^3) + 6\sqrt{x}] dx$$

$$\int x^2 \sin(4x^3) dx + \int 6x^2 \sqrt{x} dx$$

$$u = 4x^3 \quad \frac{du}{dx} = 12x^2 \quad \frac{du}{12} = x^2 dx$$

$$\frac{1}{12} \int \sin(u) du + 6 \int x^2 x^{\frac{1}{2}} dx$$

$$\frac{-\cos(4x^3)}{12} + 6 \int x^{\frac{3}{2}} dx + C$$

$$\frac{-\cos(4x^3)}{12} + 12 \frac{x^{\frac{5}{2}}}{\frac{5}{2}} + C$$

$$l) \int \frac{\ln x}{x} dx - 4 \int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$$

$$u = \ln x \quad \frac{du}{dx} = \frac{1}{x} \quad du = \frac{1}{x} dx$$

$$\int u du - 4 \int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$$

$$\frac{(\ln x)^2}{2} - 4 \int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$$

$$u = \sqrt{x} \quad 2du = \frac{1}{\sqrt{x}} dx$$

$$\frac{(\ln x)^2}{2} - 8 \int e^u du$$

$$\frac{(\ln x)^2}{2} - 8 e^{\sqrt{x}} + C$$

$$n) \int \frac{dx}{6\sqrt[3]{x^2} \cdot (36 + \sqrt[3]{x})}$$

$$\int \frac{dx}{6x^{2/3} \cdot (36 + x^{1/3})}$$

$$u = 36 + x^{1/3} \quad \frac{du}{dx} = \frac{1}{3} x^{-2/3} \quad 3du = x^{-2/3} dx$$

$$\int \frac{3x^{2/3} du}{6x^{2/3} \cdot u} \quad dx = 3x^{2/3} du$$

$$\frac{1}{2} \int \frac{1}{u} du$$

$$\frac{1}{2} \ln|u| + C$$

$$\frac{1}{2} \ln|36 + \sqrt[3]{x}| + C$$

$$n) \int \left(\sec(4x) - \frac{e^{\frac{1}{x}} + 8}{x^2} \right) dx$$

$$\int \sec(4x) dx - \int \frac{e^{\frac{1}{x}} + 8}{x^2} dx$$

$$u = 4x \quad \frac{du}{dx} = 4 \quad \frac{du}{4} = dx$$

$$\frac{1}{4} \int \sec(u) du - \int \frac{e^{\frac{1}{x}} + 8}{x^2} dx$$

$$\frac{1}{4} \ln |\sec(4x) + \tan(4x)| - \int \frac{e^{\frac{1}{x}}}{x^2} dx - 8 \int \frac{1}{x^2} dx$$

$$u = \frac{1}{x} \quad \frac{du}{dx} = -\frac{1}{x^2} \quad -du = \frac{1}{x^2} dx$$

$$\frac{1}{4} \ln |\sec(4x) + \tan(4x)| + \int e^u \cdot du + 8 \int 1 \cdot du$$

$$\frac{1}{4} \ln |\sec(4x) + \tan(4x)| + e^{\frac{1}{x}} + \frac{8}{x} + C$$

$$e) \int \frac{\ln(x)}{x} dx - 4 \int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$$

$$u = \ln(x) \quad \frac{du}{dx} = \frac{1}{x} \quad du = \frac{1}{x} dx$$

$$\int u du - 4 \int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$$

$$\frac{(\ln x)^2}{2} - 4 \int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx + C$$

$$u = \sqrt{x} \quad \frac{du}{dx} = \frac{1}{2\sqrt{x}} \quad du = \frac{1}{2\sqrt{x}} dx$$

$$\frac{(\ln x)^2}{2} - 8 \int e^u du + C$$

$$\frac{(\ln x)^2}{2} - 8e^{\sqrt{x}} + C$$

$$p) 2 \int \frac{\sin(x)}{\cos(x)} dx - 6 \int \frac{\cos(x)}{\cos(x)} dx$$

$$2 \int \tan(x) dx - 6 \int 1 dx$$

$$2 \cdot \ln|\sec(x)| - 6x + C$$

$$q) 9 \int \frac{x e^{\sqrt{3x^2+2}}}{\sqrt{3x^2+2}} dx$$

$$u = 3x^2 + 2 \quad \frac{du}{dx} = 6x \quad \frac{du}{6} = x dx$$

$$\frac{3}{2} \int \frac{e^{\sqrt{u}}}{\sqrt{u}} du$$

$$v = \sqrt{u} \quad \frac{dv}{du} = \frac{1}{2\sqrt{u}} \quad 2dv = \frac{1}{\sqrt{u}} du$$

$$3 \int e^v \cdot dv$$

$$3e^v + C = 3e^{\sqrt{u}} + C = 3e^{\sqrt{3x^2+2}} + C$$

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